

Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation

SIT746 Research Project — Thesis Defence

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Introduction

System Architecture

Feature Engineering & Core Method

Machine Learning Models

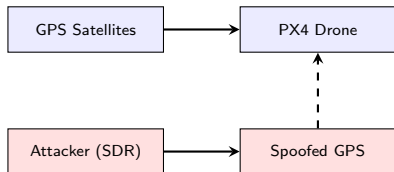
Results

GPS Trust Index

Conclusion

The GPS Spoofing Threat

- GPS spoofing: counterfeit signals with false coordinates
- **Jamming** = denial; **Spoofing** = corruption of service
- PX4 EKF2 relies heavily on GPS for state estimation
- Spoofed signals report healthy fix — attack is silent
- Low-cost SDRs $\approx$ \$300 make spoofing accessible



Real-World Incidents

- UT Austin 2012 — full 3D UAV control via spoofing^[1]
- Iran 2011 — GPS spoofing capture of RQ-170 Sentinel drone^[2]
- Commercial drone market projected \$54.6B by 2030^[3]

[1] Shepard et al., *GPS World*, 2012:

gpsworld.com/drone-hack

[2] The Aviationist, 2011:

theaviationist.com/2011/12/15/gps-spoofing

[3] Grand View Research:

grandviewresearch.com/press-release/commercial-drone-market

RQ1: GPS Vulnerabilities

What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?

RQ2: Sensor Fusion for Real-Time Detection

How can sensor fusion techniques effectively detect GPS spoofing in real time?

RQ3: GPS Trust Index

How can a dynamic GPS Trust Index be designed to quantify the reliability of GPS data?

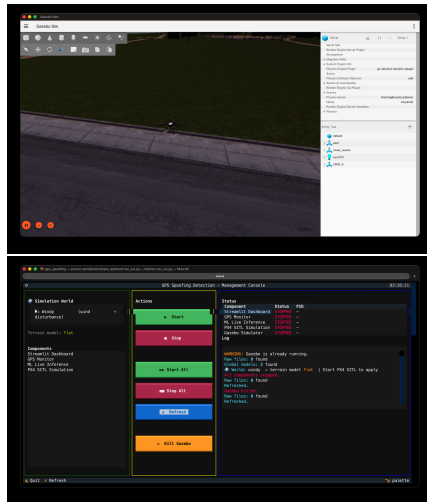
Simulation Environment & Data Pipeline

PX4 SITL + Gazebo

- Iris x500 quadcopter (3 kg)
- MAVLink telemetry @ 10 Hz
- 8 experimental + 60 synthetic flights
- 6 GPS spoofing attack modes

5-Stage ML Pipeline

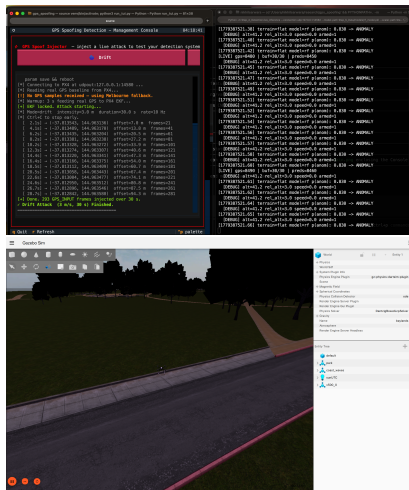
1. Clean — filter, normalize, remove gaps
2. Label — 8 physics-based heuristics
3. Window — 30 samples, 50% overlap
4. Split — 70/15/15 train/val/test
5. Train — Random Forest



GPS Spoofing Attack Injection

Attack Mechanism

- MAVLink GPS_INPUT messages @ 10 Hz
- sysid=1, compid=220 (GPS source)
- Injected on UDP port 14580
- 6 modes: Drift, Ramp, Jump+Drift, Static, Teleport, Noise



Feature Engineering — 34 Features

GPS (10)

x_m, y_m
alt_m, rel_alt_m
vel_m_s, hdg_deg
fix_type
satellites_visible
eph_m, epv_m

IMU (12)

roll_deg, pitch_deg, yaw_deg
rollspeed_radps
pitchspeed_radps
yawspeed_radps
vibration_x/y/z
clipping_0/1/2

Estimator (12)

vel_ratio, pos_horiz_ratio
pos_vert_ratio
vel_innov, pos_horiz_innov
pos_vert_innov
battery_voltage, armed
failsafe, connection_ok

Sliding windows: 30 samples $\times$ 34 features = 1,020-dimensional input
50% stride for temporal continuity — 3-second window at 10 Hz

Wind affects BOTH sensors — Spoofing affects ONLY GPS

Normal Flight

GPS: Position Changed
IMU: Motion Detected

Agree $\Rightarrow$ Normal

VS

Spoofed Flight

GPS: Position Changed
IMU: No Motion

Disagree $\Rightarrow$ SPOOFING

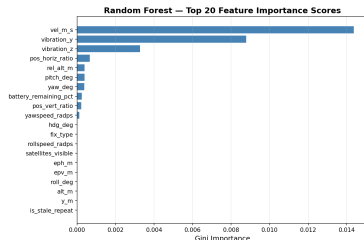
Heuristic H8: $|\omega_{\text{all}}| < 0.05 \text{ rad/s}$ and $\|\mathbf{v}_{\text{gps}}\| > 5 \text{ m/s}$ — Cannot occur under normal physics

Random Forest Classifier

Random Forest

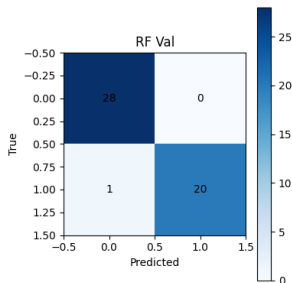
- 100 estimators, Gini impurity
- `class_weight='balanced'`
- 1,020-dim flattened input
- Feature importance built-in
- <60s training on M1 Pro, no GPU
- Interpretable, robust to class imbalance

Dataset: 11,551 windows (8,085 train / 1,732 val / 1,734 test) —
93% normal / 7% spoofed



Model Performance Results

Random Forest		
	Normal	Spoofed
Recall	0.99	0.60
F1-score	0.99	0.63

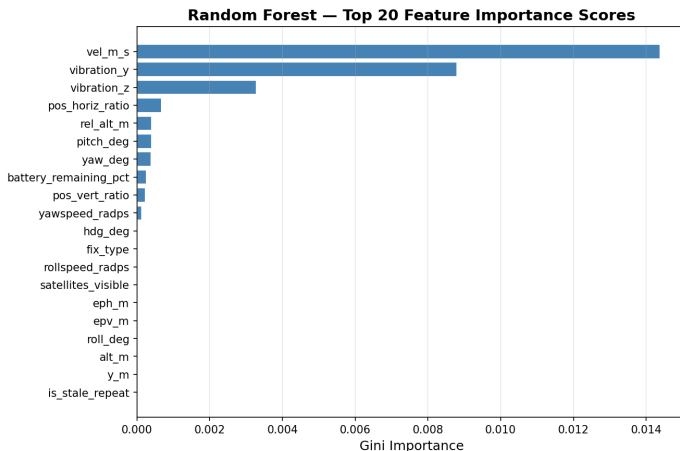


Area-Specific (Baylands) Model

- Global RF spoofed F1: 0.63
- **Area RF spoofed F1: 0.98**
- 153 real + 337 synthetic windows
- Accuracy: 0.99, Normal F1: 0.99
- Auto-activates in Baylands bounding box

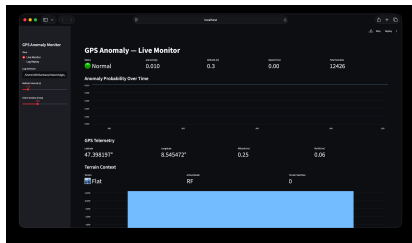
Spoofed-class detection: 18/30 test windows (RF), 0.98 F1 (area-specific)

Random Forest Feature Importance

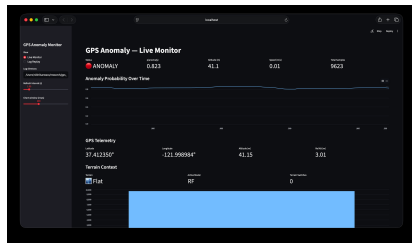


- GPS velocity and IMU angular rates are most discriminative
- EKF innovation ratios capture estimator disagreement
- Feature importance aids interpretability for operational deployment

Live Inference Dashboard



Normal flight — $GTI \approx 1.0$, $p(\text{anom}) < 0.1$

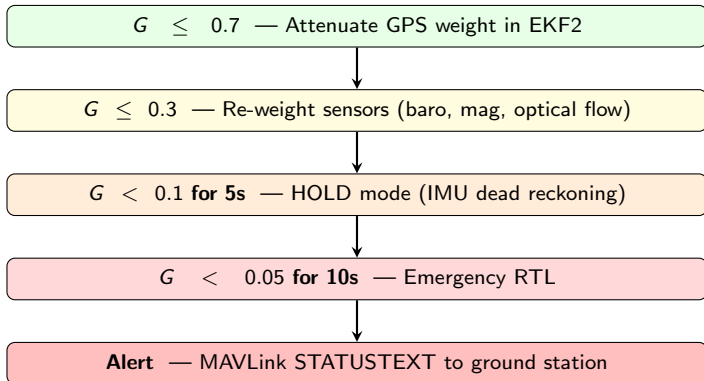


Spoofing detected — GTI drops, ANOMALY

GTI Mathematical Definition

$$G_t = \alpha G_{t-1} + (1 - \alpha)[1 - \sigma(k \cdot \Delta\Phi_t - \theta)]$$

$\alpha = 0.95$: temporal smoothing | σ : maps inconsistency $\rightarrow$ trust $\in [0, 1]$



GTI Worked Example — Normal Flight

Setup

Initial trust G_0	1.0
α	0.95
k, θ	10, 2
$\Delta\Phi$ (GPS-IMU agree)	0.1

Calculation

Step 1 — Suspicion score:

$$\sigma(10 \times 0.1 - 2) = \sigma(-1) = 0.27$$

Step 2 — Flip to trust reading:

$$1 - 0.27 = 0.73$$

Step 3 — Blend with memory:

$$G_t = 0.95 \times 1.0 + 0.05 \times 0.73 = \mathbf{0.987}$$

Result

GPS and IMU agree $\Rightarrow \Delta\Phi$ stays low $\Rightarrow$ trust remains near **1.0**.

Drone operates normally. No alerts triggered.

Key insight

$\alpha = 0.95$ acts as a memory buffer.

One turbulent reading does **not** crash the score —

only *sustained* disagreement causes trust to fall.

GTI Worked Example — Spoofing Attack

Setup

Attacker injects 50m north offset.
GPS says drone moved — IMU says it didn't.

Previous trust G_{t-1}	0.987
$\Delta\Phi$ (GPS-IMU disagree)	0.8

Calculation

Step 1 — Suspicion score:

$$\sigma(10 \times 0.8 - 2) = \sigma(6) = 0.998$$

Step 2 — Flip to trust reading:

$$1 - 0.998 = 0.002$$

Step 3 — Blend with memory:

$$G_t = 0.95 \times 0.987 + 0.05 \times 0.002 = \mathbf{0.938}$$

Trust Decay Under Sustained Spoofing

Second	Trust G	Status
0	1.000	Normal
1	0.938	Normal
5	0.723	Attenuate GPS
8	0.468	Re-weight sensors
12	0.198	HOLD mode
16	0.051	Emergency RTL

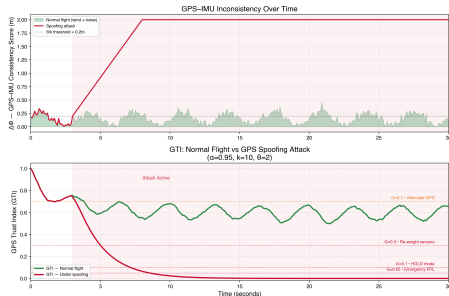
Key Insight

Gradual decay prevents false alarms from turbulence — only **sustained** spoofing triggers RTL.

GTI Analysis — Normal Flight vs Spoofing

GTI Trajectory Comparison

- **Normal flight** — GTI stays > 0.95 despite wind gusts and sensor noise
- **Spoofing attack** — GTI decays through all 5 escalation zones
- Attack detected within 1.5 s of onset
- First threshold ($G \leq 0.7$) at $t \approx 4.5$ s
- Emergency RTL at $t \approx 15.5$ s



Parameter Configuration

$\alpha = 0.95$ (slow decay, tolerates turbulence)

$k = 10, \theta = 2$ (sigmoid scaling)

$\Delta\Phi$: 0.1 m normal $\rightarrow$ 2.0 m spoofed

Contributions

1. GPS-IMU consistency discriminator (physics-based, model-free)
2. 8 heuristic auto-labelling rules
3. Complete end-to-end detection pipeline
4. Area-specific model (0.98 spoofed F1)
5. GPS Trust Index + graduated response protocol
6. Open-source, reproducible simulation environment

Future Work

1. Physical hardware validation (Pixhawk + SDR spoofer)
2. Breaking point — systematic param sweep
3. GTI integration into PX4 EKF2 firmware
4. Reduced-latency architectures (RNN/attention)
5. Multi-drone cooperative detection

RQ1: GPS Vulnerabilities

EKF2 accepts spoofed GPS due to fabricated low HDOP. Innovation gating triggers too slowly — **tens of seconds** vs milliseconds needed for safety-critical flight.

RQ2: Real-Time Detection via Sensor Fusion

GPS-IMU consistency score $\Delta\Phi$ provides the discriminative signal. RF achieves **spoofed F1 = 0.63** at 3s window; heuristic detector operates at **zero latency**.

RQ3: Dynamic GPS Trust Index

GTI (EMA-smoothed sigmoid of $\Delta\Phi$) enables a **graduated response** — from sensor re-weighting through emergency RTL — avoiding a brittle single threshold.

Detection and simulation of GPS spoofing
in mobile robots is **feasible**

using sensor cross-validation.

The GPS-IMU consistency discriminator provides a
physics-based, model-free signal that **cannot be concealed**
by any spoofing waveform.

Wind moves the drone. Spoofing moves only the coordinates.

Thank You

Questions?

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